Journal of Rock Mechanics and Geotechnical Engineering 17 (2025) 4802–4814

Contents lists available at ScienceDirect

Journal of Rock Mechanics and Geotechnical Engineering

journal homepage: www.jrmge.cn

Full Length Article

Intelligent detection method for internal fractures in mine rock masses based on borehole camera images

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ARTICLE INFO

Article history:
Received 20 April 2024
Received in revised form
18 October 2024
Accepted 21 October 2024
Available online 31 December 2024

Keywords:
Fracture detection
Borehole camera images
Convolutional neural networks (CNNs)
Attention mechanism

ABSTRACT

It is important to understand the development of joints and fractures in rock masses to ensure drilling stability and blasting effectiveness. Traditional manual observation techniques for identifying and extracting fracture characteristics have been proven to be inefficient and prone to subjective interpretation. Moreover, conventional image processing algorithms and classical deep learning models often encounter difficulties in accurately identifying fracture areas, resulting in unclear contours. This study proposes an intelligent method for detecting internal fractures in mine rock masses to address these challenges. The proposed approach captures a nodal fracture map within the targeted blast area and integrates channel and spatial attention mechanisms into the ResUnet (RU) model. The channel attention mechanism dynamically recalibrates the importance of each feature channel, and the spatial attention mechanism enhances feature representation in key areas while minimizing background noise, thus improving segmentation accuracy. A dynamic serpentine convolution module is also introduced that adaptively adjusts the shape and orientation of the convolution kernel based on the local structure of the input feature map. Furthermore, this method enables the automatic extraction and quantification of borehole nodal fracture information by fitting sinusoidal curves to the boundaries of the fracture contours using the least squares method. In comparison to other advanced deep learning models, our enhanced RU demonstrates superior performance across evaluation metrics, including accuracy, pixel accuracy (PA), and intersection over union (IoU). Unlike traditional manual extraction methods, our intelligent detection approach provides considerable time and cost savings, with an average error rate of approximately 4%. This approach has the potential to greatly improve the efficiency of geological surveys of borehole fractures.

1. Introduction

In large-scale metal mining projects, deep-hole blasting is commonly used to meet the demands of heavy equipment and high-stage mining (Ke et al., 2022). However, this method frequently leads to hole collapse and blockage near the blast zone, disrupting continuous production and endangering operator safety (Miao et al., 2015). Therefore, understanding the characteristics of joint and fracture development in rock masses is essential to ensure drilling stability in mining operations.

With the rapid advancement of mine digitization and intelligence, the limitations of traditional manual observation for fracture identification are becoming increasingly apparent, including low efficiency, high misjudgment rates, labor intensity, limited real-time capability, and reliance on subjective experience. This method is increasingly unable to meet the efficiency requirements of modern mining operations. The use of borehole camera technology to capture internal images has stimulated research interest in fracture detection through image processing, deep learning, and computer vision, both domestically and internationally. In recent years, numerous scholars have developed algorithms tailored to their specific fields and focus areas to accurately and rapidly extract fractures from images (Wang et al., 2016, 2021; Liu et al., 2019).

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Peer review under responsibility of Institute of Rock and Soil Mechanics, Chinese Academy of Sciences.

https://doi.org/10.1016/j.jrmge.2024.10.027
1674-7755/

Threshold segmentation for fracture image processing divides the original image into fracture and background regions using a threshold value T applied to pixel points in the grayscale image. Li and Liu (2008) proposed a threshold segmentation method based on domain difference histograms and formulated an objective function to maximize the difference between the two regions and effectively isolate the fracture region. Dinh et al. (2016) denoised fracture images and used the maximum peak and valley values of the grayscale histogram as segmentation thresholds. Lu et al. (2017) improved peak threshold selection by incorporating enhancement, smoothing, and denoising techniques before iterative threshold selection, enabling real-time and stable automatic threshold determination. Although threshold segmentation is computationally simple and fast, it is prone to noise, and determining the optimal threshold can be challenging. Fracture images are particularly sensitive to lighting variations during capture, often resulting in minimal pixel value differences between fractures and their surroundings, leading to suboptimal processing outcomes.

Fracture image processing using edge detection exploits significant gradient changes in pixel values along fracture boundaries, typically caused by depth discontinuities and pronounced brightness variations between the fracture and the background. This method enables effective segmentation of the fracture region from the surrounding area. Compared to threshold segmentation, edge detection significantly reduces computational requirements while preserving the structural properties of the fracture more comprehensively. Yang and Geng (2018) first preprocessed the fracture images using a nonlinear gray transformation and a reconstruction filter to enhance the linear characteristics of the fractures. They then developed an adaptive thresholding method based on the gray gradient characteristics of fractures to coarsely extract fracture edges. Candidate edge points were identified using gradient information, followed by edge processing using single-pixel percolation to exploit local pixel differences within a defined region. The final step involved filling in the edges to complete the fracture representation. Othman et al. (2019) employed an edge detection operator within the threshold range of the Canny algorithm to eliminate noise and detect fracture edges effectively. Xu et al. (2021) proposed an enhanced edge detection operator based on the Laplace operator after assessing the performance of existing edge detection methods for fracture detection. While both threshold segmentation and edge detection methods offer computational efficiency and cost-effectiveness, they often fail to detect fractures in in situ borehole images of mine rock bodies due to complex features, including intricate background textures, diverse noise levels, irregular fracture distributions, highly variable surface morphologies, and complex contact modes.

The continuous advancement of deep learning has made supervised learning-based fracture detection algorithms a research hotspot in recent years. In this context, data features are learned automatically through various implicit layers during training, eliminating the need for predefined feature extraction steps. A sufficiently large dataset is essential for training a robust detection model. Neural network-based fracture image processing excels in detecting fractures amidst complex background textures, significant noise, and diverse morphologies, accurately capturing both location information and segmentation images. Zhang et al. (2017) employed a sequence prediction model for pixel-level category segmentation of three–dimensional (3D) road cracks and developed a sequence generation model to identify the optimal local paths for fracture formation. Zou et al. (2019) performed semantic-level fracture image recognition, where the direct output of convolutional processing is the recognition result map, enabling end-to-end crack recognition. Asadi et al. (2021) introduced a multi-classifier system using deep convolutional neural networks (CNNs) to predict pixel-level fractures in rock images. Deep learning methods demonstrate powerful feature extraction and target recognition capabilities, leading to enhanced recognition accuracy, algorithmic robustness, and generalization (Rosenberger et al., 2023; Yu et al., 2024).

Despite the advancements in widely used neural network models, their performance still falls short in challenging scenarios, such as drillhole fractures with complex backgrounds and unique tubular structures. First, these models often struggle to efficiently identify and prioritize key fracture regions, leading to inaccuracies in fracture location and extent, especially when faced with complex spatial distributions and variable morphologies. Second, their inadequate capture and representation of tubular structure properties hinder understanding of subtle changes and complex fracture morphologies, making it difficult to accurately delineate complete fracture boundaries. This can result in ambiguous segmentation outcomes that prevent effective analysis of overall fracture morphology and evolutionary trends. Notably, classic models such as ResUnet (RU), VggUnet, DeepLabV3+, and SegFormer lack specialized solutions to these challenges.

To address these challenges, this study proposes an intelligent detection method for internal fractures in mine rock bodies using borehole photographic images. This approach involves generating a dataset of nodal fracture unfolding maps derived from these images and enhancing the RU model with channel attention mechanism (CAM) and spatial attention mechanism (SAM). This enhancement allows for dynamic adjustment of feature importance across image channels and optimizes spatial attention. Additionally, the method incorporates a dynamic serpentine convolution module to adaptively capture the tubular structure of fractures, refine the segmentation of borehole nodal fractures, and fit segmented fracture boundaries into trigonometric images using the least squares method. It also captures fracture inclination angle parameters. This comprehensive approach enables rapid and accurate characterization of nodal fracture evolution in rock bodies.

2. Data acquisition

2.1. Project overview

This study focuses on the Qidashan open-pit iron ore mine located in Anshan City, Liaoning Province, China, with geographical coordinates ranging from 123°05′46″ to 123°06′65″E and 41°08′20″ to 41°10′09″N. The deposit is located in the northern part of the mining belt in the Anshan area and extends for approximately 4900 m with a dip angle exceeding 70°. It features a northwest strike and dips either southwest or northeast, supporting a production scale of 17 million tons annually. The exposed rock formations on the quarry slope mainly consist of old granite with most of the joints and fractures filled by closed veins of quartz and calcite. The fault surfaces and rock groups display a rough texture, and the rock body is characterized by polygonal blocks. Fig. 1 illustrates the geological map of the Qidashan deposit.

The selected experimental blasting area has a step elevation ranging from −120 m to −105 m, with a section height of 15 m and a blasting step length of 132 m. The step width varies between 20 m and 28 m, and the slope angle is set at 65°. The step area measures 3168 m^2, with a total volume of 47,520 m^3 and a step rock quantity of 104,664 tons. The experimental blasting involves slag blasting utilizing 62 holes arranged in a rectangular layout, as shown in Fig. 2a. The hole spacing is 7.5 m, and the row spacing is 6 m, with detonation occurring sequentially from hole to hole. Each hole has a diameter of 250 mm and a depth of 17.5 m, corresponding to a stage height of 15 m.

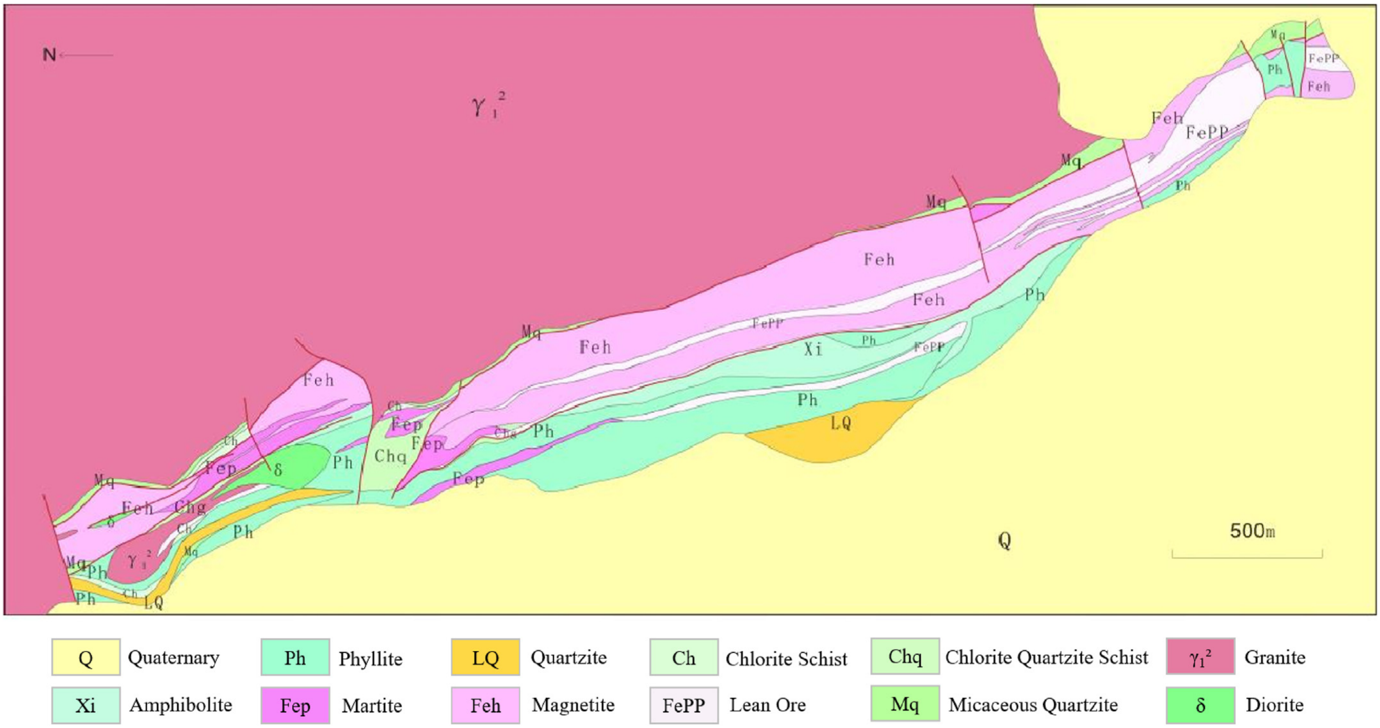
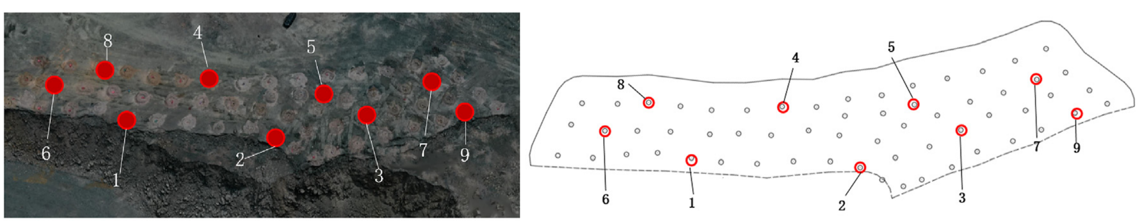


Fig. 1. Geological map of the Qidashan deposit.



(a) (b)

Fig. 2. (a) Bench and (b) borehole locations for data collection.

2.2. Borehole imaging acquisition

Currently, borehole camera technology (Zou et al., 2018) is widely utilized in various fields, including glacier engineering, geotechnical engineering, mining engineering, and engineering geology, to address the challenges associated with large burial depths. In this study, we used the CU301 digital panoramic borehole camera system developed by the Wuhan Geotechnical Research Institute of the Chinese Academy of Sciences. This system integrates electronic, video, digital, and computer application technologies and features an innovative optical structure design that overcomes the limitations of precious analog drilling imaging technologies to achieve panoramic and digital imaging capabilities.

A borehole camera was employed for image acquisition in this study, as shown in Fig. 3. A three-person team worked collaboratively: one focused on geological information acquisition and data processing, while the other two ensured that the probe moved towards the borehole at a steady uniform speed. They recorded video of the borehole wall throughout the probing process.

Ten drillholes were strategically designed for data collection in the experimental blasting area, arranged in three rows in a rectangular formation to ensure a comprehensive representation of joints and fractures across the entire terrace. Due to the presence of groundwater in certain boreholes, data acquisition depth was limited to 12 m in each borehole. Ultimately, nine boreholes, numbered 1–9, were successfully acquired and analyzed. The specific locations of the selected boreholes are illustrated in Fig. 2b.

Fig. 3. Data acquisition process using panoramic camera instruments.

2.3. Borehole imaging processing

The initial data are collected from the photographic video within the borehole. The digital panoramic drilling camera system generates extensive data from the borehole video images, which appear as circular inverted trapezoids, making them unsuitable for human visual observation and difficult to analyze using traditional image processing methods. As a result, supporting software is essential for the automatic recognition and storage of depth data from the digital panoramic borehole camera, as well as for integrating electronic compass data and generating annular and scan-line images from the borehole images (Wang et al., 2017a).

Panoramic camera probes utilize a unique optical transformation with conical mirrors to superimpose orientation information on a 360° image of the borehole wall, resulting in a flat or panoramic image. However, the annular panoramic image generated through this optical transformation suffers from distortion, making direct observation difficult. Digitizing the panoramic image establishes a conversion relationship between the original borehole images and panoramic images. This process allows synchronized display of the panoramic image, the planar unfolding image, or the virtual drill core image. The planar unfolding image presents a two-dimensional (2D) view of the entire (360°) borehole wall, as if it had been vertically split along the north pole and unfolded into a planar surface (Han et al., 2015; Wang et al., 2017b). The virtual drill core map provides a three-dimensional (3D) representation created by rewinding the planar unfolding, allowing observation from an external viewpoint, as shown in Fig. 4.

2.4. Dataset generation

A valid dataset is essential for training predictions using network models. Dataset generation involves standardizing image dimensions and labeling borehole images to provide training data for the image segmentation network. Here, borehole images were acquired at a depth of approximately 12 m. However, soil covering the borehole interior necessitated flushing to obtain clearer images, making data collection challenging. Ultimately, the study compiled 54 sheets containing 65 sets of fracture data, all uniformly cropped to a resolution of 1007 × 6700 pixels. To address the issue of insufficient sample data, we applied data augmentation techniques while preserving the intrinsic characteristics of the borehole fractures. Techniques such as flipping, rotating 180°, and flipping with rotation were used to expand the dataset to a total of 216 sheets. A geologist carefully annotated the borehole images to ensure accurate delineation of fracture contours. Fig. 5 displays locally annotated images adjusted in three different ways.

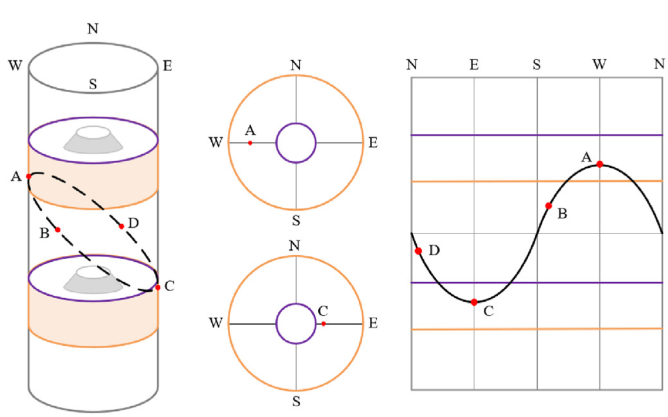


Fig. 4. Schematic diagram of imaging principle.

The creation of the borehole photographic image dataset involves a sequential process: acquisition of borehole photographic videos, pre-processing of panoramic images into planar unfolded maps, and annotation of the data. Although data augmentation helps to ensure the reliability of the dataset, it remains relatively small and inadequate for training deep or wide networks, which have numerous parameters that can lead to overfitting. Additionally, the dataset features a complex image background, with a low percentage of irregularly distributed fracture pixels, characterized by tubular structures and varying morphologies. This study aims to develop segmentation networks with multiscale detection capabilities, attention mechanisms, and robust feature extraction abilities. These enhancements will improve the effectiveness of the model in localizing fractures and performing fine segmentation in complex backgrounds.

3. Methods

The proposed method for the intelligent detection of internal fractures in mine rock bodies primarily utilizes artificial neural networks for fracture segmentation. Given the simplicity of image semantics and the small dataset size, the network model is designed to be relatively small with fewer parameters to prevent overfitting. Consequently, the Unet model, widely used for semantic segmentation in medical applications (Ronneberger et al., 2015), is adopted. The basic Unet framework is enhanced by integrating the Resnet50 backbone, resulting in the RU network model. Additionally, the convolutional block attention module (CBAM) is incorporated into the RU model. The CAM dynamically adjusts the importance of features across channels, improving feature diversity and accuracy. The SAM optimizes the model's focus on critical regions of the image, enhancing target positioning and segmentation accuracy. Furthermore, the dynamic snake convolution (DSC) module allows for adaptability in capturing tubular fracture structures and finer details through dynamic convolution operations, improving the model's recognition and segmentation abilities, and overall robustness. This approach enables high-quality pixel-level segmentation, even with a limited number of training samples.

3.1. RU

Resnet (He et al., 2016), or residual network, is a deep CNN that utilizes residual connections to facilitate more efficient training of deep architectures. These residual connections implement a cross-layer connection method, enhancing the network's ability to learn low-frequency information effectively. In contrast, Unet is a fully convolutional network specifically designed for image segmentation tasks. It features an encoder–decoder structure that integrates both global and local context information, addressing the challenge of missing semantic information in segmented outputs.

Building upon this, RU (Diakogiannis et al., 2020) combines the residual connections of Resnet with the Unet architecture, incorporating multiple residual connections within each encoder and decoder module. The shortcut connections, also known as skip connections, established in each residual block create cross-layer interactions, as shown in Fig. 6. This design helps to mitigate the gradient vanishing problem often encountered in deep networks. Additionally, the residual blocks enhance the efficient extraction of low-frequency information, while the fusion of contextual information effectively addresses the challenge of missing semantic information. As a result, RU demonstrates a significant improvement in image segmentation accuracy.

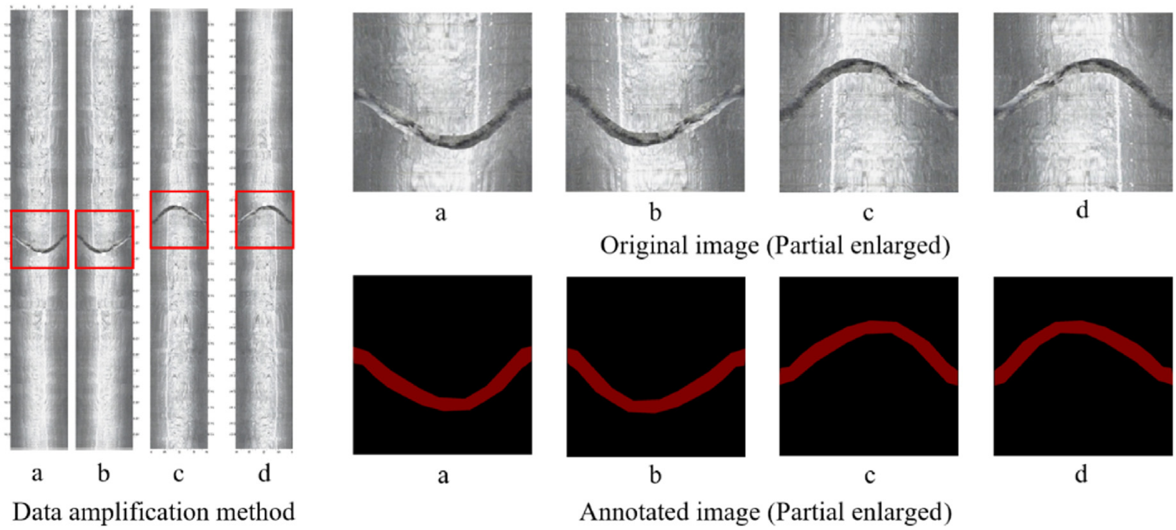


Fig. 5. Data amplification and annotated images (partial enlarged): (a) Original image; (b) Flip image; (c) Image rotated by 180°; and (d) Flip with image rotated by 180°.

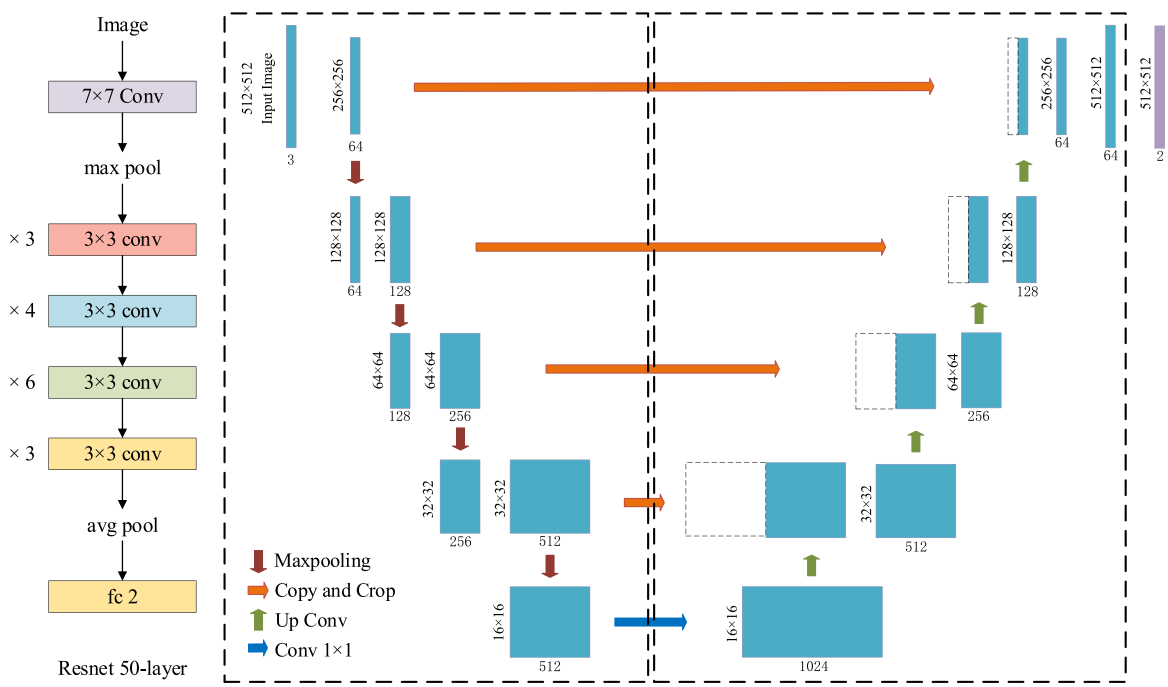


Fig. 6. RU network structure.

3.2. RU-CBAM

Convolutional attention, specifically CBAM (Woo et al., 2018), is designed for image classification and target detection by integrating CAM and SAM. This combination allows for adaptive weighting of feature maps, enhancing the network's representation and generalization capabilities. The CBAM consists of two key components: CAM and SAM. CAM dynamically reassigns weights to each channel in the feature map to emphasize important features while suppressing irrelevant ones. This is achieved using the concept of self-attention, where the feature map is first compressed into a single vector through global average pooling. The weights for each channel are then calculated using fully connected layers. Finally, these weights are multiplied with the original feature map, enhancing the representation of significant features. In contrast, SAM focuses on different regions of the image to improve local feature extraction. It generates a spatial attention map through convolutional operations, which emphasizes noteworthy regions in the feature map. Specifically, two parallel 3×3 convolutional layers are applied to process the width and height dimensions of the input feature map. The results are then fused to create a 2D attention map, which is applied to the feature map to highlight key regions in a weighted manner.

(1) CAM

The input feature map F, with dimensions $H \times W \times C$, undergoes global max pooling and global average pooling, producing two

$1 \times 1 \times C$ feature maps. These maps are processed individually through a two-layer neural network (multi-layer perceptron, MLP). The first layer has C/r neurons (where r is the reduction rate) and utilizes the ReLU activation function. The second layer consists of C neurons. Notably, this two-layer neural network shares weights. After the MLP processes both feature maps, the outputs are summed element-wise and passed through a sigmoid activation, resulting in the final channel attention feature M_c. Finally, an element-wise multiplication is performed between M_c and the input feature map F to assign the corresponding attention weights to each channel, generating the input features for SAM:

$$\begin{aligned} M_c(F) &= \sigma(MLP(AvgPool(F)) + MLP(MaxPool(F))) \\ &= \sigma\left(W_1\left(W_0\left(F^c_{avg}\right)\right) + W_1\left(W_0\left(F^c_{max}\right)\right)\right) \end{aligned} \tag{1}$$

where σ represents the sigmoid function, $W_0 \in R^{C/r \times C}$ performs dimension reduction operation, and $W_1 \in R^{C \times C/r}$ performs dimension increase operation. The two inputs share the MLP weights, and W_0 follows the ReLU activation function.

(2) SAM

The feature map F' produced by the CAM serves as the input feature map for this mechanism. Initially, global maximum pooling and global average pooling are applied to obtain two $H \times W \times 1$ feature maps. These maps are then concatenated (channel splicing) to create a single $H \times W \times 2$ feature map. A 7×7 convolution operation is then applied, which is considered to be more effective than a 3×3 convolution, resulting in a $H \times W \times 1$ feature map. Subsequently, a sigmoid function is applied to generate the spatial attention feature, denoted as M_s. Finally, this feature is multiplied by the input feature map to produce the final output.

SAM utilizes channel-based global average pooling and global maximum pooling operations to produce two feature maps that represent distinct information. These maps are merged and then fused through a 7×7 convolution, which features a large receptive field. Finally, a sigmoid operation generates a weight map that is overlaid onto the original input feature map, enhancing the target region. In spatial attention, the interaction between channels is neglected, as features in each channel undergo equal processing. In contrast, channel attention involves direct global processing within a channel, which may overlook spatial interaction among features:

$$\begin{aligned} mIoUM_s(F) &= \sigma\left(f^{7\times7}([AvgPool(F); MaxPool(F)])\right) \\ &= \sigma\left(f^{7\times7}\left(\left[F^s_{avg}; F^s_{max}\right]\right)\right) \end{aligned} \tag{2}$$

where $f^{7\times7}$ represents a convolution operation with a filter size of 7×7.

Integrating the CBAM into the RU encoder structure involves placing it after each residual block. The channel attention component learns relationships between channels by calculating the mean and maximum values for each channel, resulting in channel attention weights. These generated weights are then applied to each channel's feature map, adjusting its significance accordingly. The spatial attention component learns relationships between different spatial locations by calculating the mean and maximum values for each location, generating spatial attention weights. The resulting weights are applied to the entire feature map, thereby adjusting the importance of spatial locations. By incorporating the CBAM into the RU encoder structure, the network gains the ability to learn both channels and spatial relationships within the input data. This flexibility enhances the model's perception of essential features.

Schematic diagram of each attention submodule, as shown in Fig. 7. The channel submodule combines both maximally pooled outputs and average pooled outputs within a shared network. Simultaneously, the spatial submodule utilizes two similar outputs pooled along the channel axis and forwards them to the convolutional layer.

3.3. RU-DSC

Accurate extraction of borehole fractures encounters challenges, such as elongated and fragile local structures, as well as complex and variable global morphology. The model may struggle to perceive these structures due to the small proportion of borehole fractures in the image, limited pixel composition, and interference from complex backgrounds. This can cause the convolution kernel to stray from the target, making it difficult to accurately discriminate small changes, ultimately resulting in fragmentation and breakage in the segmentation results. DSC (Qi et al., 2023) addresses these challenges by considering the slender and continuous serpentine morphology of borehole fractures. The DSC module combines traditional convolution with a serpentine path to efficiently capture multiscale features. Unlike standard convolution, dynamic serpentine convolution moves along the serpentine path of the feature map, allowing the network to capture a larger range of spatial relationships while maintaining computational efficiency. The dynamic adjustment of the serpentine path is achieved through parameterization, granting the network flexibility to modify the size and shape of the convolution kernel based on input feature characteristics, as shown in Fig. 8. This approach enhances the perception of serpentine structures, ultimately improving segmentation performance for borehole fractures.

$$K_{i\pm c} = \begin{cases} (x_{i+c}, y_{i+c}) = \left(x_{i+c}, y_i + \sum_i^{i+c} \Delta y\right) \\ (x_{i-c}, y_{i-c}) = \left(x_{i-c}, y_i + \sum_{i-c}^{i} \Delta y\right) \end{cases} \tag{3}$$

$$K_{j\pm c} = \begin{cases} \left(x_{j+c}, y_{j+c}\right) = \left(x_j + \sum_j^{j+c} \Delta x, y_j + c\right) \\ \left(x_{j-c}, y_{j-c}\right) = \left(x_j + \sum_{j-c}^{j} \Delta x, y_j - c\right) \end{cases} \tag{4}$$

where $K_{i\pm c}$ and $K_{j\pm c}$ represent the specific positions of each grid in the convolutional kernel; c denotes the horizontal distance from the center grid; and Δx and Δy signify the offsets applied to K_{i+1} relative to K_i, achieved through bilinear interpolation.

3.4. Fracture feature extraction method

Feature extraction of the fracture will be based on the segmentation results. The observed rock structure surface in the drilling image can be divided into two parts: the upper and lower boundaries of the borehole fracture contour. Ideally, these two contour lines should have similar morphology. After pixel-level segmentation of the borehole fracture using our proposed model, extracting the pixel point coordinates of the boundary line becomes straightforward. As the structural surface fracture in the unfolded panoramic borehole image resembles a characteristic sinusoidal curve, we select the sinusoidal curve as the characteristic function. We then determine a set of parameters to minimize the sum of squares of residuals between the points on the sinusoidal curve and

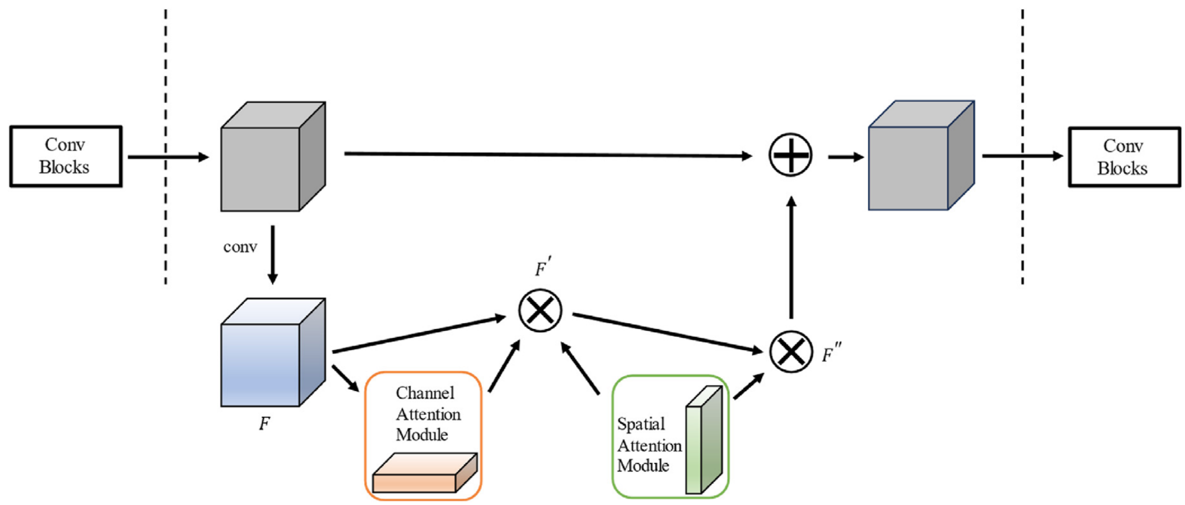


Fig. 7. CBAM in the encoder structure of RU.

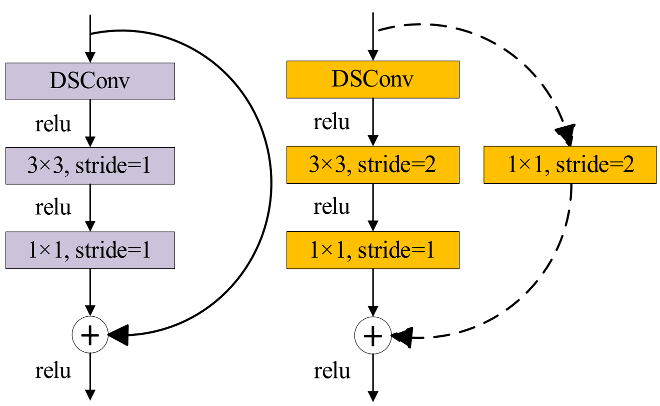


Fig. 8. DSC module in the encoder structure of RU.

those on the contour lines, employing the least squares method to fit the optimal sinusoidal curves of the contour lines, as illustrated in Fig. 9a.

After fitting the upper and lower contour lines, the horizontal and vertical coordinates of the borehole image are converted into pixel point coordinates with a consistent scale. This facilitates the calculation of the structural surface's base plane and essential parameters, including the depth of the corresponding structural surface based on the optimal sinusoidal curve's position, the tendency of the structural surface based on the phase angle of the curve, and the inclination based on the magnitude of the curve (Zou et al., 2020). The results are illustrated in Fig. 9b. The analytical equation for the adopted sinusoidal function is provided as follows:

$$f(x) = A\sin(wx + \varphi) + h \tag{5}$$

where $A = (y_A - y_B)/2$, $w = \pi/(x_A - x_B)$, $\varphi = \pi(x_A + x_B)/[2(x_A - x_B)]$, and $h = (y_A + y_B)/2$.

The highest and lowest intersection points of the two parallel planes of the fracture opening and the borehole are A, A', and B, B', respectively. The structural plane dip angle $\alpha = \arctan[(y_A - y_B)/(2r)]$, the inclination angle $\beta = \varphi - \pi/2$ when $\varphi \in (\pi/2, \pi)$ and $\beta = \pi - \varphi$ when $\varphi \in (-\pi, \pi/2)$, and the aperture $d = AA' \cos\alpha$.

4. Results and analysis

4.1. Model training

The experiment was conducted on a computer platform running Windows 10, equipped with an Intel (R) Core (TM) i7-11700 processor, an RTX 3090 graphics card, and 32 GB of RAM. The network model was built using the Python 3.9 interpreter, leveraging GPU acceleration to enhance segmentation accuracy and speed. To ensure a balanced evaluation of network performance, a combination of data from various regions and boreholes was utilized before the division into training and test datasets, with 80% allocated for training and 20% for testing. The model training employed

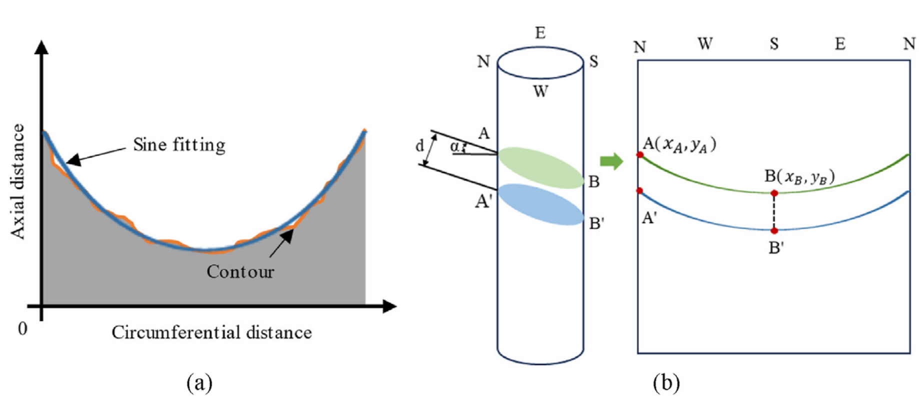


Fig. 9. Trigonometric function fitting and coordinate relationship transformation: (a) Sine curve fitting fracture contour line; and (b) Transformation of coordinate relations.

the Adam optimizer, with an initial learning rate of 0.001, and performance was evaluated using the cross-entropy function. The batch size was set to 8, while the other parameter settings can be found in Table 1.

To ensure a fair comparison between the improved RU network model proposed in this paper and the original RU, as well as VggUnet (Wang and Hu, 2021), DeepLabV3+ (Chen et al., 2018), and SegFormer (Xie et al., 2021), we employed the same initialization method, loss function, and optimization algorithm parameters across all models. During pretraining experiments, the model was saved after 300 iterations. Upon analysis of the loss values and accuracy curves, it was observed that after approximately 230 iterations, the training dataset's accuracy increased slowly, while the testing dataset's accuracy rose more rapidly. This indicated that the network had reached sufficient training. To enhance network performance and mitigate overfitting, we decided to terminate training after 250 iterations, as shown in Fig. 10.

4.2. Evaluation metrics

The proposed algorithm in this paper is evaluated using three widely accepted measures for image segmentation (Wang et al., 2020; Ma et al., 2023a):

(1) Accuracy (global pixel accuracy). This metric represents the total number of correctly classified pixels as a percentage of the total number of pixels, providing an overall assessment of the model's performance across the entire image.
(2) PA (pixel accuracy). This measures the average proportion of correctly classified pixels within the fracture category, reflecting the model's accuracy specifically within that category.
(3) IoU (intersection over union ratio). Highly sensitive to the accuracy of the model's segmentation boundaries for fracture categories, this metric quantifies the degree of overlap between the model's predicted regions and the true labeled regions for fractures (Ma et al., 2023b).

All of these metrics range from 0 to 1, with values closer to 1 indicating better segmentation performance by the model.

4.3. Segmentation results and analysis

4.3.1. Model comparison

A series of experiments were conducted to evaluate the effectiveness of the proposed RU model in comparison to RU, VggUnet, DeepLabV3+, and SegFormer. The variation curves for accuracy, PA, and IoU for each model are illustrated in Fig. 11. All five models demonstrate stable convergence in the metrics while achieving pixel-level accurate segmentation of borehole fractures. The improved RU proposed in this paper outperforms the other four models in terms of accuracy, PA, and IoU, yielding segmented fractures that are closer to the labeled images. The remaining models, in order of their superiority, are RU, VggUnet, DeepLabV3+, and SegFormer.

During the initial training stages, the accuracy of the five models fluctuated significantly on the test dataset before ultimately exceeding 99%. This high accuracy rate can be attributed to the imbalance in pixel proportions between the borehole fracture and the background, with fracture pixels occupying a much smaller percentage. In terms of evaluation metrics, PA and IoU for the fracture category reveal that our model significantly outperforms the other four, especially in IoU, where it exceeds 90% (Table 2). This indicates the model's effectiveness in achieving accurate pixel-level segmentation of borehole fractures. Additionally, Fig. 12 shows the segmentation results for each model. Both DeepLab and SegFormer struggle with accurately segmenting fracture contours, while RU exhibits edge failures at the fracture's ends but performs well in the middle inflection positions. VggUnet, although better at the ends, struggles in the middle. In contrast, our model, which employs an attention mechanism and a dynamic serpentine convolution module, accurately detects the fracture shape and effectively segments its edges against the background.

Based on these results, the improved network proposed in this paper is expected to perform better in the segmentation of new

Table 1
Hyperparameter setting.

input_shape	batch size	init_lr	momentum	optimizer_type	lr_decay_type	dice_loss
512 × 512	8	0.0001	0.9	Adam	Cos	True

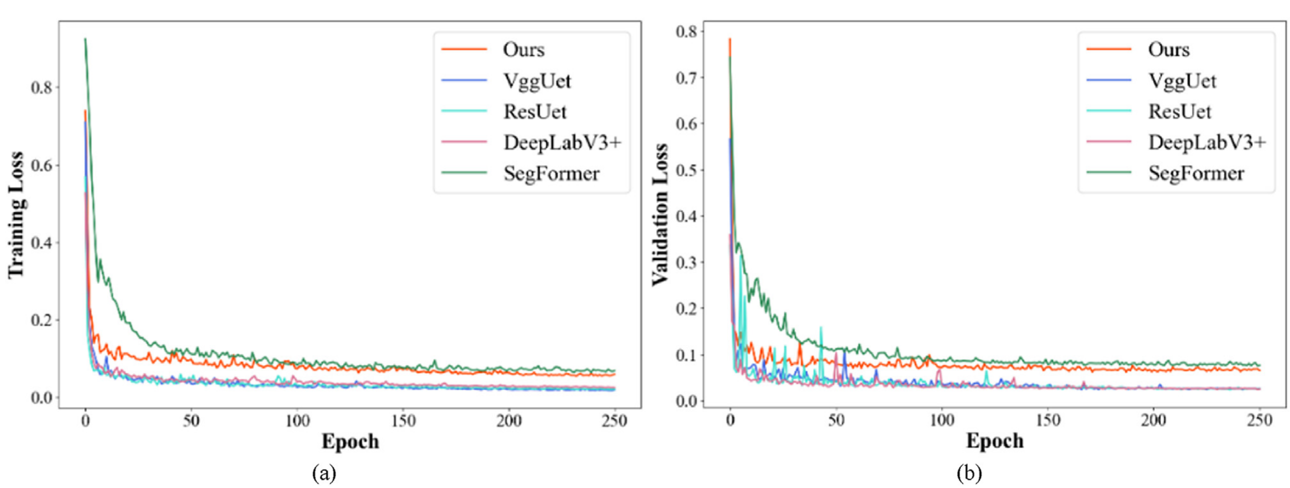


Fig. 10. Loss value change curve during model training and verification: (a) Training Loss; and (b) Validation loss.

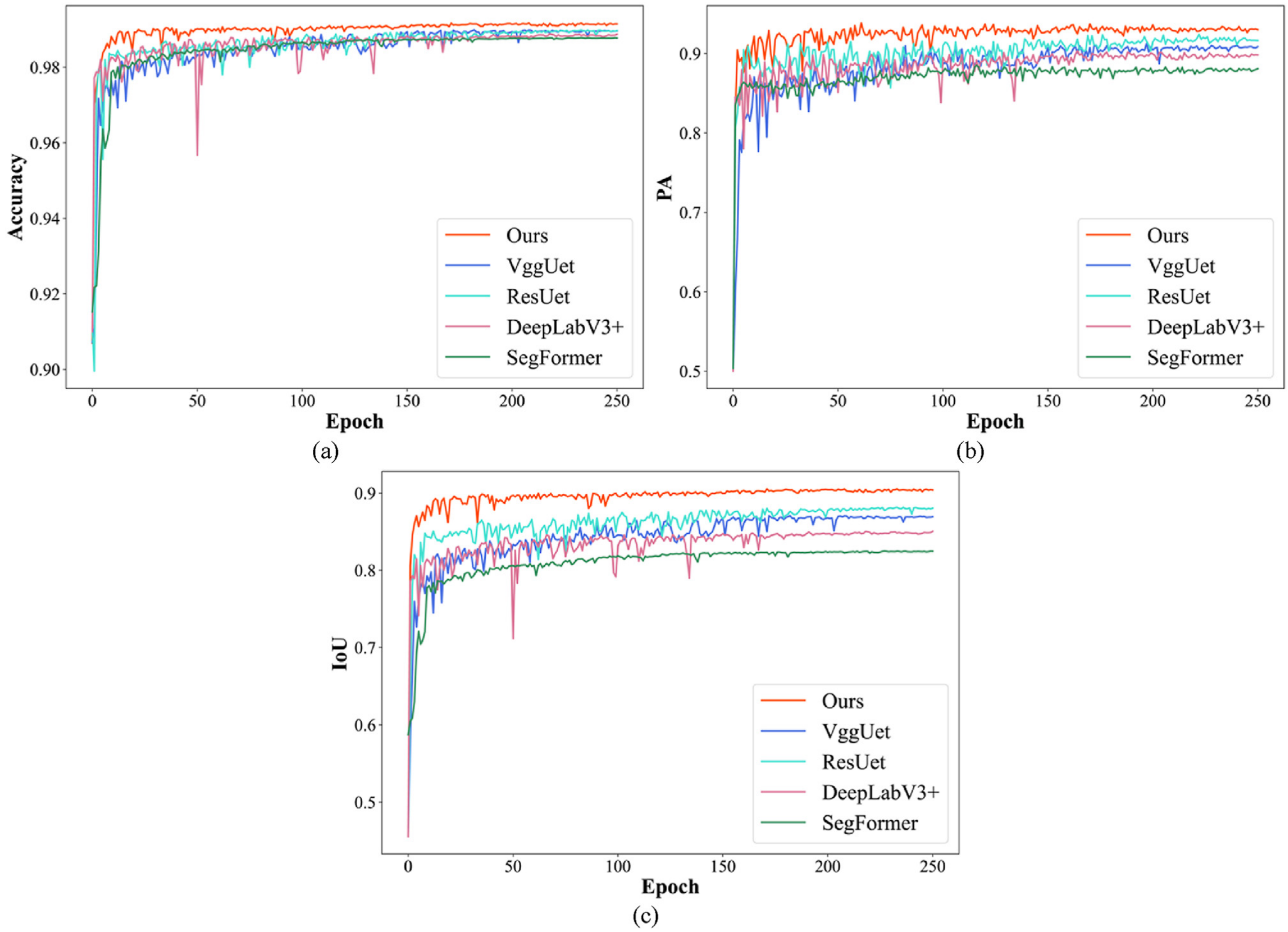


Fig. 11. Curves of various evaluation indicators: (a) Accuracy, (b) PA; and (c) IoU.

images, thereby solidifying its status as a highly effective segmentation network for borehole images.

4.3.2. Ablation experiment

In this study, we designed and integrated both the CBAM and DSC modules into the RU architecture to enhance the network's segmentation capability for borehole fractures. The CBAM focuses the model's attention on the fracture regions while minimizing background interference, thereby improving segmentation accuracy. However, further refinement is required for edge extraction of fractures. Meanwhile, the DSC module adaptively modifies the convolution kernel shape to better recognize serpentine structures, effectively addressing inaccuracies due to blurred boundaries between fracture edges and the background. Using consistent experimental parameters, we conducted ablation experiments to validate the efficacy of these modules in our proposed method (Zhao et al., 2023). As shown in Fig. 13, the symbol “N” indicates the absence of the module, while “Y” denotes its inclusion.

Table 2
Quantitative extraction results of the borehole fracture.

Model	Accuracy	PA	IoU
Proposed	0.992	0.931	0.904
VggUnet	0.989	0.907	0.869
RU	0.989	0.916	0.879
DeepLabV3+	0.988	0.898	0.848
SegFormer	0.987	0.881	0.824

The experimental results indicate that both the CBAM and DSC modules significantly enhance network segmentation. Although adding both CBAM and DSC modules increases the model size by 12.01%, both PA and IoU metrics for the fracture category show marked improvement compared to configurations without these modules or with only one of them. Additionally, there was no substantial increase in computation time or resource requirements during the experiments. This suggests that the enhanced method proposed in this paper is effective and exhibits better generalization ability.

4.3.3. Explanation through visualization

While the improved RU model's superior performance has been validated through multidimensional evaluation metrics, it is essential to gain a deeper understanding of how the deep learning architecture focuses on target features. This analysis will elucidate the intrinsic working mechanism of the model and confirm the specific contributions of the newly introduced CBAM and DSC modules to its performance enhancement. Recent studies have utilized class activation maps and heat map visualization to reveal the activation patterns of CNNs in key regions during image processing. Among these techniques, Grad-CAM offers an intuitive visual explanation of the decision-making processes within deep CNNs (Gao and Mosalam, 2018; Pope et al., 2019; Selvaraju et al., 2020).

In this study, we utilized heat map visualization to compare and analyze the weighted integration properties of the feature maps generated by the modified RU model and the original RU model at

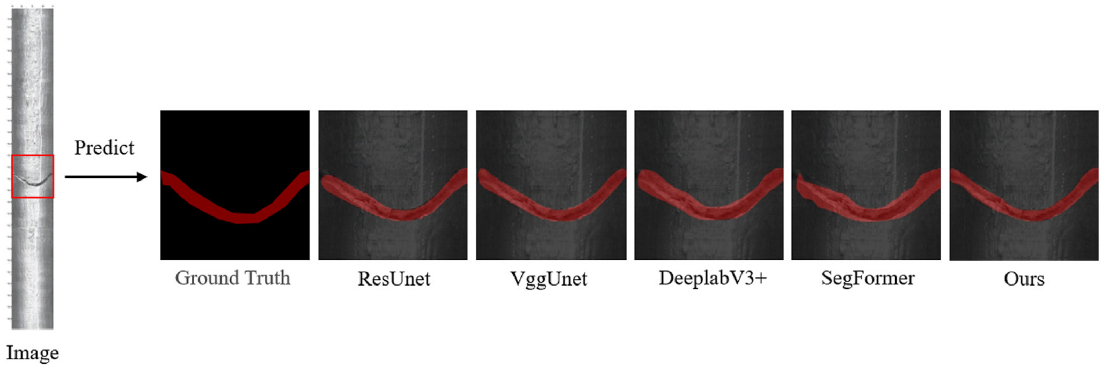
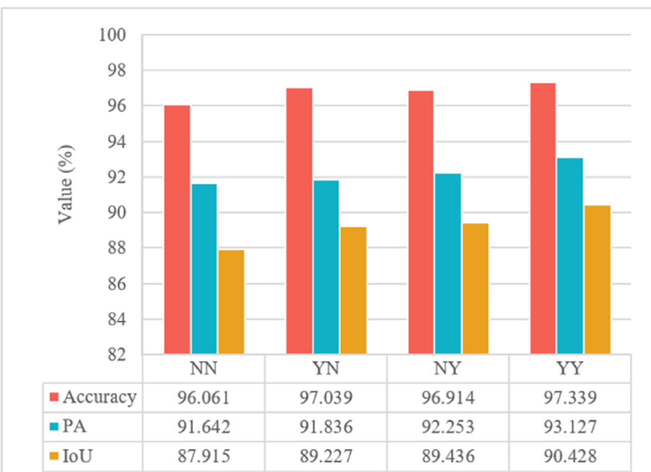


Fig. 12. Visual comparison of borehole fracture image segmentation results.



	NN	YN	NY	YY
Accuracy	96.061	97.039	96.914	97.339
PA	91.642	91.836	92.253	93.127
IoU	87.915	89.227	89.436	90.428

Fig. 13. Ablation experiments of different modules.

each convolutional stage. At the low-level convolutional layers (Fig. 14b), the improved RU model demonstrates a strong ability to recognize the fracture morphology, while the original RU model only identifies the general location of the fracture. In the mid-level network (Fig. 14c), both models effectively capture the main region of the target fracture and its boundary morphology. However, the improved RU shows greater sensitivity to edge detection, focusing more on the fracture edges. When examining the feature response in the deep structure (Fig. 14d), the improved RU, enhanced by the CBAM and DSC modules, significantly focuses on the intact fracture region, aligning closely with human expert recognition during high-level feature extraction. In contrast, the original RU tends to focus on the core region and broader sections of the fracture, neglecting the subtle edges and narrower regions.

In summary, the visual evidence presented in Fig. 14 strongly supports the beneficial impact of the proposed CBAM and DSC modules on the RU model. These modules enhance the model's ability to refine features that characterize the target fracture while effectively guiding it to prioritize and strengthen the core and detail regions of the fracture, leading to improved overall performance. Consequently, further extraction of fracture production parameters

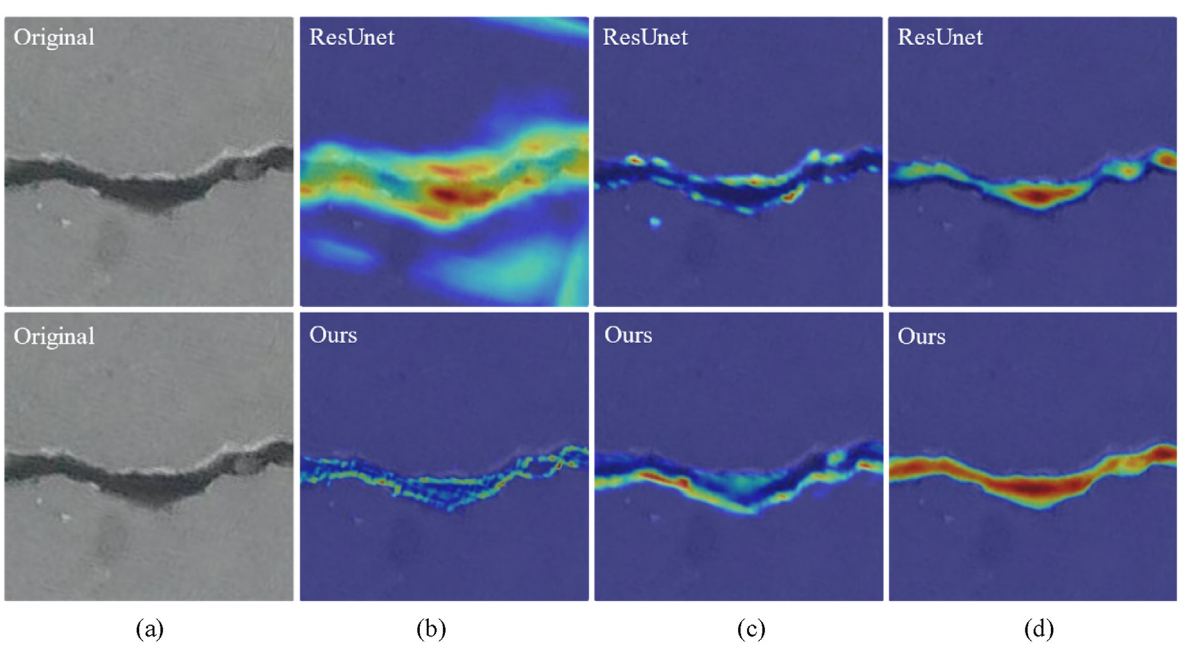


Fig. 14. Heat maps from different convolutional layers of two models: (a) Original images, (b) Heat maps of the low layer, (c) Heat maps of the middle layer and (d) Heat maps of the deep layer.

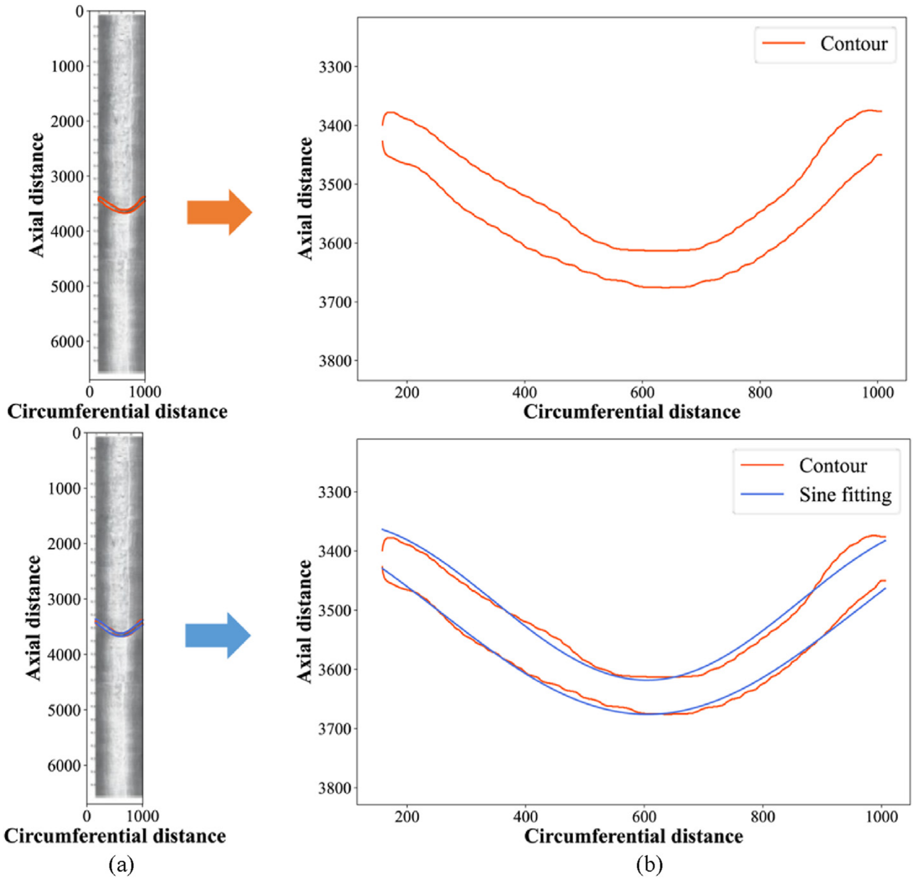


Fig. 15. Fitting the upper and lower boundaries of borehole fracture contours using sine curves: (a) Extraction of upper and lower boundaries of borehole fracture contours; and (b) Sinusoidal curve fitting.

using the enhanced RU model proves to be more effective, and optimizing the model by increasing the variety and quantity of sample data is also valuable.

4.3.4. Extraction results and analysis

Fig. 15a presents a segmented image of a borehole fracture produced by the modified RU model, showing the extracted upper and lower boundaries of the contour, which represent the edge points of the segmented fracture image. Fig. 15b illustrates the fitting of these boundaries using sinusoidal curves through the least squares method. By correlating the intrinsic lateral resolution of the drilling equipment with the borehole diameter, we can map the positional information of the contour points to their pixel positions in both horizontal and vertical coordinates. Table 3 summarizes the quantitative extraction results of the borehole fracture parameters, including depth (h), dip angle (α), inclination (β), maximum gap width (d), and average gap width ($\overline{d}$).

Moreover, the results of manual interpretation, following thorough proofreading, are considered accurate standard data. An error margin of less than 15% for the extraction results is defined as the criterion for accurately identifying the structural surface. Based on this standard, identification segmentation and quantitative extraction of each parameter are conducted on the validation set, achieving an approximate accuracy of 83.5% in the quantitative extraction of borehole fracture yield. The average error in structural surface parameters is approximately 4.3%. Observations indicate that misclassifications and missed structural facets predominantly occur in regions of the rock body that are structurally fractured, especially where the fractures exhibit irregular patterns. The imaging quality of the borehole wall in these regions is subpar, complicating the interpretation of rock structure fragmentation features. Conversely, in regions with higher imaging quality and more apparent structural surfaces, the recognition segmentation and quantitative extraction method presented in this paper effectively segments the structural surface and accurately extracts its characteristic parameters, showing minimal error when compared to manual recognition.

By training the deep learning model to segment rock images, the location, morphology, and other features of the nodal fractures can be automatically recognized, allowing for the calculation of their production parameters. The proposed method for the intelligent detection of internal fractures in mine rock masses, utilizing borehole photographic images, is both feasible and reliable. This approach significantly enhances working efficiency, reducing task durations from several days to just a few hours, thereby expediting project progress and conserving both human and material resources.

Table 3
Quantitative extraction results of borehole fracture.

h (m)	α (°)	β (°)	d (mm)	$\overline{d}$ (mm)
−15.23	15.71	81.57	3.81	3.52

5. Conclusions

An intelligent detection method was proposed in this study for internal fractures in mine rock bodies using images obtained from borehole photography technology. This method includes a fracture segmentation method and a feature extraction method for the segmented fractures. The fracture segmentation method addresses challenges such as inaccurate identification of fracture regions and unclear detection of fracture contour edges. By replacing Resnet50 with the Unet network model as the backbone and incorporating the CBAM attention mechanism module and the DSC dynamic serpentine convolution module, an improved RU network model was established. This model extracts fracture parameters, such as inclination, dip, and gap width, using the least squares method to fit the segmented image using a sinusoidal curve. The improved RU model outperforms RU, VggUnet, DeepLabV3+, and SegFormer on the established borehole fracture image dataset. These improvements were validated through ablation experiments and model heat map visualizations. Furthermore, the proposed feature extraction method significantly reduces the time required compared to traditional manual measurements and achieves an average error of around 4% in parameter extraction. Thus, the proposed intelligent detection method effectively replaces manual labor, reduces costs, and increases the efficiency of geological surveys concerning borehole fractures.

The future work should focus on the following aspects:

(1) Geological conditions and borehole morphology vary across different mines. The weights of the improved network trained on the Qidashan mine dataset can be utilized as pretrained weights for training on other mine datasets using transfer learning, thereby accelerating model convergence.
(2) Given the time-consuming nature of creating labeled datasets, we aim to develop a high-quality dataset encompassing various fracture morphologies to ensure sample diversity and representativeness for improved training. Additionally, we aim to integrate this method into the host computer of borehole camera instruments for real-time image processing and analysis, further improving work efficiency.

CRediT authorship contribution statement

Xinbo Ma: Writing – original draft, Methodology, Conceptualization. **Fuming Qu:** Supervision, Project administration, Methodology, Formal analysis. **Wenxuan He:** Investigation, Data curation. **Liancheng Wang:** Validation. **Xiaobo Liu:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was financially supported by the National Natural Science Foundation of China (No. 52474172).

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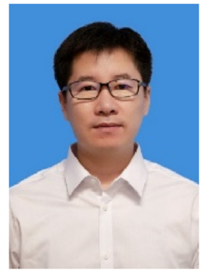

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